



# Technical Data & Specification Sheet

Smart Panel in 50 mm, 75 mm, 100 mm and 120 mm – lightweight, insulated and earthquake-ready.

## Product Overview

Smart Sandwich Panel: a light EPS-and-cement core wrapped in 4.5 mm non-asbestos calcium silicate board on both faces, with a tongue-and-groove edge so every panel locks into the next.

Smart Solid Panel: a lightweight interlocking panel made from EPS beads, cement, sand, water and bonding agents, reinforced with mesh. Both sides are plastered after installation.

## Sizes

| Thickness | Length Options (mm) | Width (mm) |
|-----------|---------------------|------------|
| 50 mm     | 2400 / 2700 / 3000  | 610        |
| 75 mm     | 2400 / 2700 / 3000  | 610        |
| 100 mm    | 2400 / 2700 / 3000  | 610        |
| 120 mm    | 2400                | 610        |

## Technical Data – Sandwich Panel (thickness in mm)

| Technical Detail                     | 50    | 75    | 100   | 120   |
|--------------------------------------|-------|-------|-------|-------|
| Anti-Impact Performance (times)      | ≥8    | ≥10   | ≥15   | ≥15   |
| Anti-Bending Damage (× dead weight)  | ≥3    | ≥3    | ≥3    | ≥3    |
| Anti-Pressure Strength (MPa)         | ≥5    | ≥5    | ≥5    | ≥5    |
| Softening Coefficient                | ≥0.80 | ≥0.80 | ≥0.80 | ≥0.80 |
| Surface Density (kg/m <sup>3</sup> ) | ≤45   | ≤61   | ≤78   | ≤78   |
| Moisture Rate (%)                    | ≤10   | ≤9    | ≤7    | ≤7    |
| Drying Shrinkage Value (mm)          | ≤0.45 | ≤0.5  | ≤0.6  | ≤0.6  |
| Single Point Hanging Strength (N)    | ≥1200 | ≥1300 | ≥1500 | ≥1500 |
| Sound Insulation Capacity (dB)       | ≥35   | ≥40   | ≥45   | ≥45   |
| Fire Proof Time Limit (h)            | ≥3    | ≥4    | ≥4    | ≥4    |
| Thermal Conductivity (W/m·K)         | ≥0.30 | ≥0.30 | ≥0.30 | ≥0.30 |

## Salient Features

- Thermal Resistant / Insulation
- Sound Resistant
- Fire Resistant
- Cost Saving
- Water Resistant
- Quick Installation
- Space Saving
- Earthquake Resistant

## Why Builders Choose Smart Panel

### Seismic Safety

A lighter wall means less force hits the structure when the ground shakes: lower base shear, less deflection.

### Speed & Lower Cost

Tongue-and-groove panels click into place fast, cutting labor time and bringing handover forward.

### Certified & Approved

ISO 9001:2015 certified company. Listed on the Government District Rate List.

### Green Construction

No topsoil dug up, no kiln smoke released, and built in part from recycled glass fiber.

## Applications

### Commercial & High-Rise

Lighter walls trim structural steel and foundation costs by 5-6%.

### Healthcare & Education

Quiet, fire-resistant spaces suited to learning and recovery.

### Industrial & Agriculture

Fast build-out with steady indoor temperatures year-round.

### Residential Builders

Earthquake-ready homes, cooler in summer and warmer in winter.

### Note

Technical values are manufacturer-stated figures for Smart Sandwich Panel as given in the company profile. For project-specific design, structural calculations and test certificates, please contact the Smart Panel technical team.